

PROJECT SPECIFIC RAM PLAN - IPS SOLUTION

Digital Interlocking ÖBB

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LOG OF CHANGES			
Revision	Date	Author	Change Description
01 PD 01	2025-09-26	Florian Moser	First proposal
01 RL	2025-10-03	Florian Moser	See 1.5.2

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1. SCOPE

1.1 PURPOSE

This document presents the Reliability, Availability and Maintainability Plan (RAM) Integrated Platform Services (IPS) solution element – IPS Solution OBB 1.0.0 (IPSS OBB 1.0.0).

The RAM plan defines the reliability, availability and maintainability objectives, tasks to be performed and documents to be produced and delivered at various phases of the project's lifecycle, to address the functional and technical requirements specified in the respective feature lists and requirements documentation.

For information about organisation, work packages, milestones and application of the Management System refer to [SEMP_IPS].

1.2 IDENTIFICATION

Project Name:	DSTW ÖBB Entwicklung
Project ID:	100025795 / ATRELDE91
Solution / System:	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
Solution Element:	Solution Element – IPS Solution OBB 1.0.0 (IPSS OBB 1.0.0)
Contract:	Digitales Stellwerk DSTW – Rahmenvereinbarung
Identification¹:	IPS 40.0.0 ADM 40.0.0 HW GENCAB SERV 1.0 HW GENCAB NETW 1.0
Customer:	Generic Application Digital Interlocking OBB (GA DIL OBB 1.0.0)
End Customer:	ÖBB
SIL Level:	NON-SIL / Non-safety solution

Table 1: Solution Identification

Security Level:	SL-T 3, SL-C 3 ²
------------------------	-----------------------------

¹ The identification refers to the key products/solutions included in the Solution Element - IPS Solution

² On the SL please note that ÖBB and also EULYNX clearly asks for SL3, an end-to-end implementation in Phase 1 will most likely NOT be possible. However, the setup and the describing documents shall be prepared for SL3. MiniOMC will not be classified as SL3 since it is based on OMC 3.0

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Maturity Level:	ML 3
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Table 2: Security/Maturity Level

1.3 INTENDED AUDIENCE

The intended audience of the project specific RAM Plan are the following functions:

- Project Design Authority (PDA)
- Project Manager (PM)
- Solution Engineering Manager (SEM)
- RAM Assurance Manager (RAM-AM)
- Safety Assurance Manager (SAM)
- Quality Assurance Manager (QAM)
- Validator (VAL)
- Verifier (VER)
- Security Assurance Manager (SecAM)

1.4 DOCUMENT MANAGEMENT

The management of this document will be performed according to the Document Management Rules [DMR] and the Document List and Review Matrix [DLRM].

1.5 CHANGE HISTORY

1.5.1 EDITION 01 PD 01

- First proposal.

1.5.2 EDITION 01 RL

- Review comments considered.

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2. REFERENCES AND TERMS

2.1 REFERENCED DOCUMENTS

Id	Reference
[ARC_GA]	Top Level Architecture Drawing - Generic Application - Digital Interlocking OBB; 3BU 17300 1218 DSAPA, Ed.02
[CRS_DSTW]	DSTW Streckenausrüstung OBB - Teil D Leistungsbeschreibung und Lastenhefte; 3BU 17300 0004 DSAPC, Ed.04
[PH_RAMs]	Pflichtenheft Digital Interlocking OBB - D2.2 Teilpflichtenheft RAMS, 3BU 17300 0022 DTAPC, Ed.02
[SD_DSTW]	Digitales Stellwerk - Technische Konzepte 3BU 17300 0010 DSAPC, Ed.04
[RAM_Plan_Generic]	RAM Plan Generic; 3BU 15000 1712 DUAPA, Ed.06
[RAMP_DFE]	Project specific RAM Plan – DSTW - Digital Field Elements; 3BU 62800 0018 DUAPA, latest
[RAMP_DIL]	Project specific RAM Plan – DIL – Generic Application / Solution Element Virtual Interlocking; 3BU 17300 1209 DUAPA, latest
[RMP_IPS]	Requirements Management Plan (RMP) Solution Element IPS; 3BU 17300 2010 DUAPA, latest
[SEMP_GA]	System Engineering Management Plan (SEMP) Generic Application Digital Interlocking OBB; 3BU 17300 1200 DPAPA, latest
[SEMP_VIL]	System Engineering Management Plan (SEMP) Solution Element Virtual Interlocking; 3BU 15000 6955 DPAPA, latest
[SEMP_IPS]	System Engineering Management Plan (SEMP) IPS Solution - Digital Interlocking OBB; 3BU 17300 2001 DPAPA, latest
[SEMP_DFE]	SEMP Solution Element - Digital Field Elements; 3BU 62800 0000 DPAPA, latest
[PBS_DSTW]	Product Breakdown Structure (PBS) - Generic Application - Digital Interlocking OBB; 3BU 17300 1215 DPAPA, Ed.02
[SSS_HW]	Hardware SSS / Generic Server and Network Cabinet; 3BU 16000 0002 DTAPA, Ed.04
[SSS_IPS]	Subsystem Requirements Specification; Integrated Platform Services (IPS); 3BU 17300 2051 DTAPA, Ed.01PD02
[RMP_GA]	Requirements Management Plan (RMP) Generic Application Digital Interlocking OBB; 3BU 17300 1207 DUAPA, latest

Table 3 : Referenced Documents

2.2 STANDARDS AND PROCEDURES

Id	Reference
[Alcarel]	Reliability Standard Failure Rate Factor; 1AA 00210 0001 BCZZA
[Birolini]	Reliability Engineering. Theory and Practice; ISBN: 354040287X, 6 th Ed
[DLRM]	Document List and Review Matrix for engineering projects; 83530224-DOC-AUT-EN-010
[DMR]	Document Management Rules; 83530061-DOC-AUT-EN-013
[EN 50126-1]	EN 50126-1:2017 Railway applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS), Part 1: Generic RAMS process.
[IEC60050(192)]	IEC 60050-192:2015; International Electrotechnical Vocabulary (IEV) - Part 192: Dependability

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Id	Reference
[MIL_HDBK217F]	US MIL-HDBK 217F:1991, Reliability prediction for electronic equipment

Table 4 : Standards and Procedures

2.3 TERMS, ABBREVIATIONS AND ACRONYMS

Term/Abbreviation	Meaning
ADM	Administration Diagnostic Monitoring
CE	Computing Element; One logical computing channel. Can be executed on a physical or virtual machine.
CG	Computing Group; Set of computing nodes, replicated for availability. Examples of a CG are 1oo2 CN-redundancy, 1oo4 CN-redundancy, 2oo4 CN-redundancy, but also 1oo1 CN-redundancy in the extreme case.
CN	Computing Node; Set of replicated Computing Elements building a safe entity. Examples of a CN are 2oo2 CE-redundancy, but also 1oo1 CE-redundancy in the extreme case.
DCA	Distributed Controller Architecture
DDSP	Daten Diagnose Service Plattform
DFE	Digital Field Elements
DGP	Diagnostic Processor
DPR	Dependability Requirement
DIL / DSTW	Digital Interlocking = DSTW, Digitales Stellwerk = DIL
EC-MUX	Element Controller Multiplexer
EULYNX	European Initiative to Linking Interlocking Systems
IEC	International Electrotechnical Commission
IPS	Integrated Platform Services
Failure rate	Limit for $\delta t \rightarrow 0$ of the probability that an item will fail in the time interval $(t, t + \delta t]$, given that the item was new at $t = 0$ and did not fail in the interval $(0, t]$, divided by δt . [Birolini]
FiT	Failures in time (unit: $10^{-9}h^{-1}$)
FES	Field Element Station
Ground Benign (GB)	Nonmobile, temperature and humidity-controlled environments readily accessible to maintenance; includes laboratory instruments and test equipment, medical electronic equipment, business and scientific computer complexes, and missiles and support equipment in ground silos. [MIL_HDBK217F].
Ground Fixed (GF)	Moderately controlled environments such as installation in permanent racks with adequate cooling air and possible installation in unheated buildings; includes permanent installations of air traffic control radars and communication facilities [MIL_HDBK217F].
GB 40	Ground Benign 40° C
GF 40	Ground Fixed 40° C
[MIL_HDBK217F]	The Military Handbook is published by the US Department of Defense, and provides mathematical models for electrical, electronic and electro-mechanical components. Each model calculates the component failure rate for given combinations of environmental conditions, component value and type, electrical stress conditions and manufactured quality level
MTBF	The expectation of the operating time between failures. [IEC 60050(191)]. $MTBF$ is usually reserved for items with <i>constant</i> failure rate λ . In this case, $MTBF = 1/\lambda$ is the expected value (mean) of the <i>exponentially distributed item's failure-free operating time</i> . In particular $MTBF \approx T/k$, where T is the given, <i>fixed</i> cumulative operating time and k the total number of failures (failed items) during T . [Birolini] See also [EN 50126-1].

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MTTR	The expectation of the time to restoration. [IEC 60050(191)] <i>MTTR</i> is the expected value (mean) of an item's repair time. It is obtained from the distribution function $G(t)$ of the repair time as $MTTR = \int_0^{\infty} (1 - G(t))dt$. For reliability or availability calculation a constant repair rate μ (i.e. exponentially distributed repair times with $\mu = 1/MTTR$) can often be used to get approximate results. [Birolini] See also [EN 50126-1].
NMR	NxMooM Redundancy, Examples of NMR are 2x2oo2, 4x2oo2, but also 1x1oo1 in the extreme case. NxMooM Redundancy is a shorthand for 1ooN CN-redundancy with each CN forming an MooM CE-redundancy.
Nx2oo2	N times 2 out of 2, Denotes a 1ooN CN-redundancy with each CN forming a 2oo2 CE-redundancy.
PBA	Printed Board Assembly
POD	Smallest deployable unit in Kubernetes context.
VM	Virtual Machine (OpenStack)

Table 5 : Terms, Abbreviations and Acronyms

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3. DESCRIPTION OF THE SYSTEM

For detailed information on the products used, refer to [PBS_DSTW].
For a detailed description of the IPS Solution refer to [SD_DSTW].
For an overview about customer expectation see [CRS_DSTW].
Information about Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) and Solution Element Virtual Interlocking Phase 1 (VIL OBB 1.0.0) can be found in [SEMP_GA], [SEMP_VIL] and [RAMP_DIL].
Information about Digital Field Elements (DFE) can be found in [SEMP_DFE] and [RAMP_DFE].

3.1 SYSTEM OVERVIEW

The IPS Solution is a core element of the Digital Interlocking ELEKTRA 4.0 (DSTW). It provides the virtualization infrastructure for the Virtual Interlocking (VSTW) and Administration, Diagnosis & Monitoring (ADM) subsystems, as well as COTS computing and network hardware. The system is designed to support up to 435 interlockings across geo-redundant computing centers, enabling centralized control and high availability.

The top-level architecture of the technical product "Generic Application Digital Interlocking (GA DIL)" is illustrated in Figure 2, taken from [ARC_GA].

The Integrated Platform Services (IPS) solution element provides the core virtualization platform for the Digital Interlocking Central Unit (DIL-CU), see Figure 1 below.

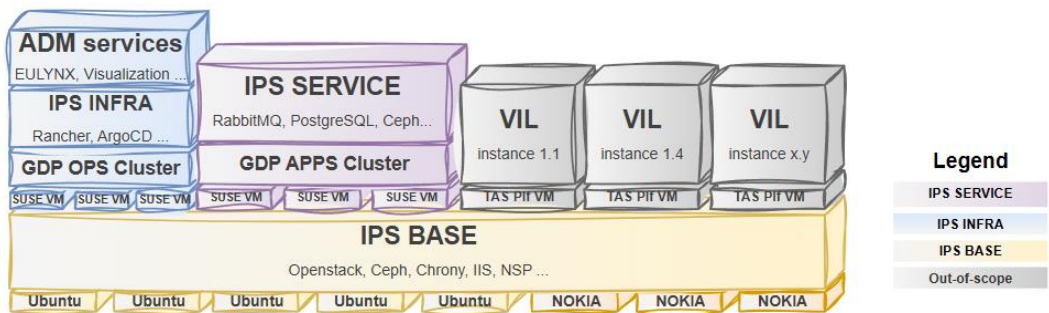


Figure 1: IPS Overview

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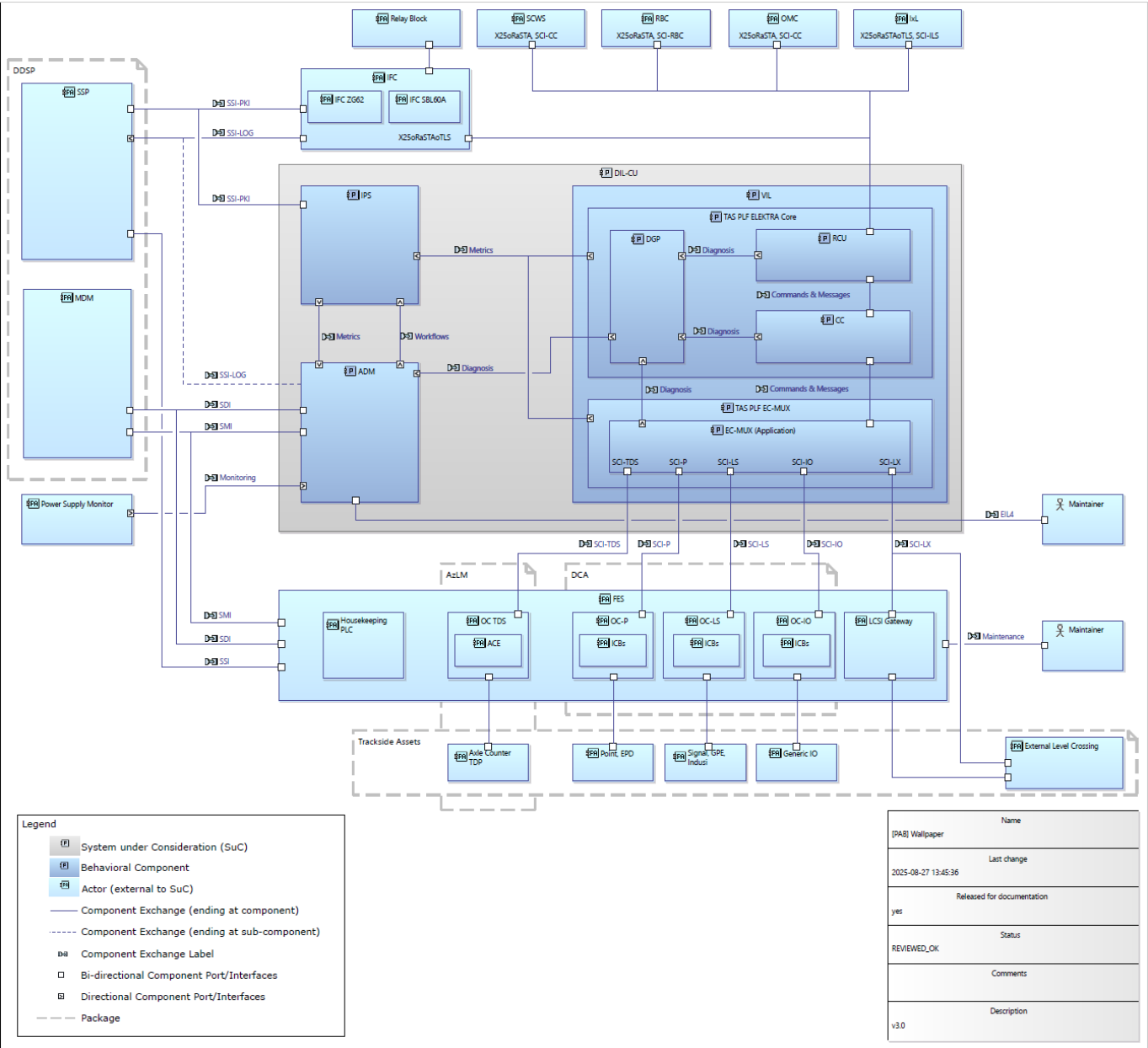


Figure 2: System architecture GA DIL

3.2 HARDWARE COMPONENTS

The IPS Solution comprises several hardware elements, structured for modularity and maintainability:

- HW GENCAB SERV 1.0: Server cabinet hosting virtualized applications.
- HW GENCAB NETW 1.0: Network cabinet for internal and external communication.

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The IPS platform operates on Standard Commercial Off The Shelf (COTS) hardware, the current plan is to use HPE ProLiant DL345 Gen 11 Servers. Nokia 7250 IXR-e routers are currently planned for the network.

- The IPS components are logically structured into three functional areas per location: Network, Computing, and Storage.
- These components are physically distributed into Availability Zones (AZs), typically corresponding to physical racks.
- The AZ serves as the largest assumed failure unit per location. If a complete rack fails (one AZ), the remaining two Availability Zones maintain sufficient computing power and storage capacity to ensure uninterrupted operation of the location.

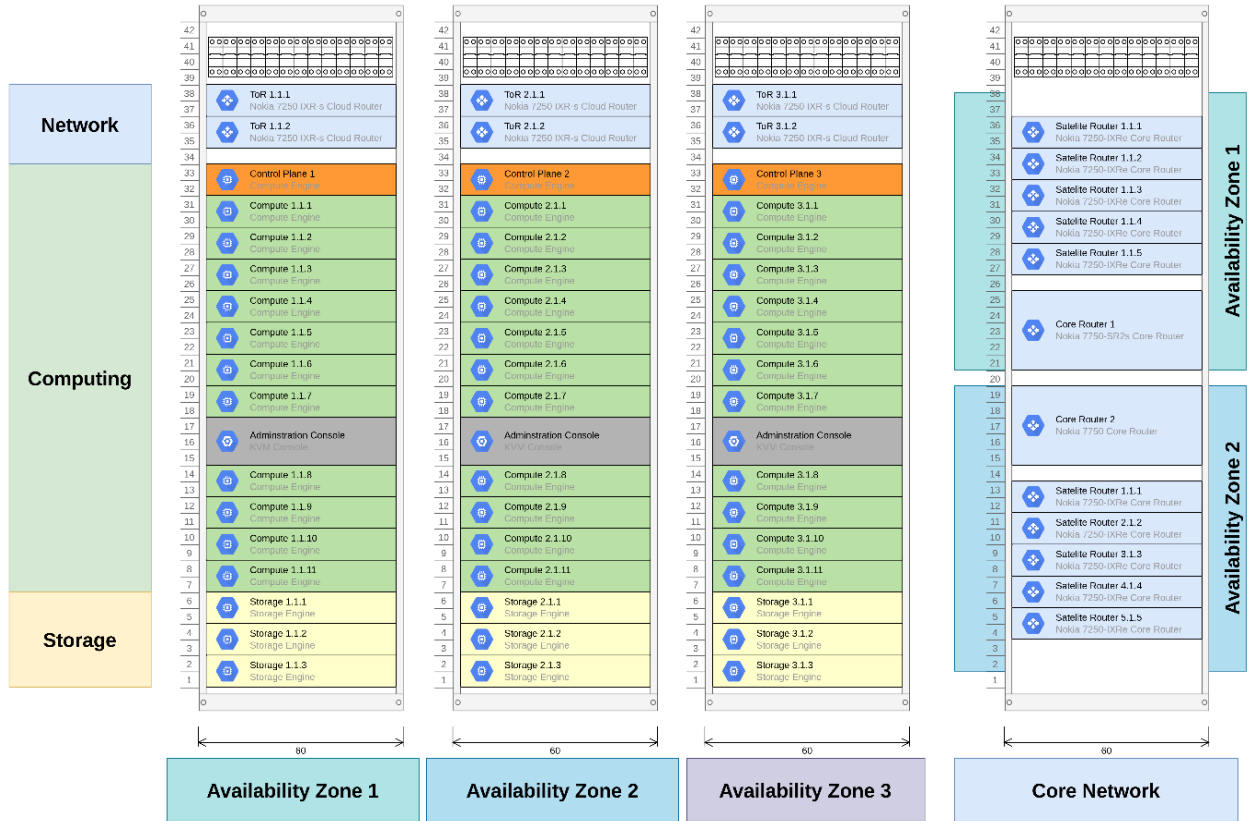


Figure 3: IPS basic structure

All hardware components, such as Computing Servers, Network Routers, and Power Distribution Units (PDUs), are designed to be redundant, both logically and physically.

Servers and network devices are equipped with at least two power supplies (PSUs) and fans, allowing for hot-pluggable component exchange.

The power supply distribution within the cabinets ensures that critical components like servers, routers, and switches have two hot-redundant (1oo2) power supplies connected to different PDUs.

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3.3 RAM DESIGN RULES

In order to ensure that the availability requirements are met, a series of rules, the so-called 'RAM Design Rules', will be developed and used as binding conditions for architectural design decisions. Note that each of these RAM Design Rules will also be explicitly mapped to one or more specific system requirements, ensuring traceability and compliance throughout the development lifecycle.

They share some common definitions described in the following table

Term	Definition	Comment
Balanced distribution	A distribution of k items to n buckets is called balanced if the number of items per bucket differs by at most 1.	This term can be used for a balanced distribution of HW servers as well as for a balanced distribution of (SW) components.
Balanced increment	An increase in a balanced distribution by one item, so that the resulting distribution remains balanced.	This term can be used for a balanced increment of HW servers as well as for a balanced increment of (SW) components.
Affinity	A principle which establishes a preference for certain elements to work together or be placed together.	Node Affinity: allows to constrain which nodes a Pod/VM can be scheduled on. Pod/VM Affinity: allows to schedule Pod/VM to be co-located with other Pod/VMs.
Anti-Affinity	A principle which ensures that certain elements are not being placed together to avoid single points of failure	Node Anti-Affinity: ensures that a Pod/VM is not scheduled on certain nodes. Pod/VM Anti-Affinity: ensures that a Pod/VM is not scheduled on the same node as certain other pods.
Equi-Distribution	Distributing workloads, data, or resources evenly to ensure balanced utilization and avoid overloading any single component	The balanced distribution defined above is an example of an equi-distribution.
Topology Spread	Distribution of elements across different availability zones	In OpenStack context: Distribution of pods across different failure domains (for example availability zones) to ensure high availability and fault tolerance.

Table 6: RAM Design Rules Related Definitions

Note:

- A **node** in the context of OpenStack refers to a physical server on which multiple virtual machines (VMs) can be operated.
- A **pod** is a term from Kubernetes and represents the smallest deployable unit (container). In contrast, in OpenStack, VMs are the smallest unit of work.

The common definitions and partitioning of the RAM Design Rules are described in detail in [RAMP_DIL] and is not in scope of this document.

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4. RAM REQUIREMENTS

The reliability, availability and maintainability requirements have to be identified and referenced within the Dependability Analysis. This Dependability Analysis should also provide a proper apportionment and refinement of these requirements according to the system design. A general description of all RAM aspects can be found in [RAM_Plan_Generic].

4.1 RELIABILITY REQUIREMENTS

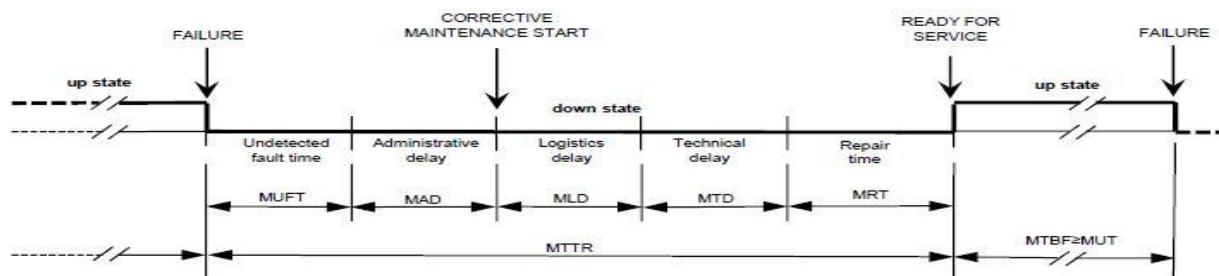
Concerning reliability, RAM activities have to focus on the following goals:

- Determination of all possible failures of the system with focus to random hardware faults. Analysis of their consequences with respect to reliability, availability and maintainability (MTTR) and calculation of the associated failure rates.
- Determination of the impact of failures on the functionality of the system.

The key parameters for describing RAM features of a system, on the basis of [EN 50126-1] are:

- Mean Up Time MUT, resp. Mean Time Between Failure MTBF,
- Mean time to Restore MTTR and the
- Availability A, resp. Unavailability ($U = 1-A$).

Figure 4, obtained from [EN 50126-1], illustrates key parameters of the availability concept.



Key

MTBF	mean (operating) time between failures
MUT	mean up time
MUFT	mean undetected fault time
MAD	mean administrative delay
MLD	mean logistics delay
MTD	mean technical delay
MRT	mean repair time
MTTR	mean time to restore (for corrective maintenance)

Figure 4: Availability concept and related terms

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The MUT is the mean lifetime of a system between two failures. MTTR gives the time required from failure to restart operation.

4.2 AVAILABILITY REQUIREMENTS

A system analysis shall be carried out to determine availability figures of components and sub-systems. The result shall be part of the Reliability Analysis. It shall be pointed out, that the availability figures depend on the proper installation, maintenance and operation conditions of the system and sub-systems as stated in the installation and maintenance manuals.

4.3 MAINTAINABILITY REQUIREMENTS

Concerning maintainability, RAM activities have to focus on the following goals:

- Identification and implementation of diagnostic measures to minimize system downtime
 - Detect fault alarms that have to be recorded during operation of the system.
 - Prepare appropriate counter measures in case of failures.
- Determine the time necessary for failure diagnosis and recovery of the availability of the system in case of system or component failures.
- Identification of necessary maintenance actions including time required to conduct them.

All requirements concerning maintenance activities need to be summarized to plan necessary actions. Those actions subsume inspection, preventive and corrective maintenance, repair and spare part logistic.

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5. RAM PLAN

5.1 METHODS AND TOOLS USED

The reliability calculation will be done according to [EN 50126-1] and the [MIL_HDBK217F] based Alcatel Reliability Standard [Alcarel] and according to the environmental conditions *Ground Benign (GB)* for indoor components and *Ground Fixed (GF)* for outdoor components.

5.2 RAM DETAILED ACTIVITIES, ANALYSES AND DOCUMENTS

Within this section, documents and activities necessary for defining and analysing reliability, availability and maintainability requirements are described.

5.3 RAM DESIGN RULES

It is planned to develop a series of rules, so-called RAM Design Rules as binding conditions for architectural design decisions if possible. Each of these RAM Design Rules will be mapped to one or more specific requirement(s). See also section 3.3 RAM Design Rules.

5.4 RAM HW-FMEA

It is recommended to create a RAM HW-FMEA, that aims to analyse availability and maintainability aspects in terms of failed HW components or failed sub-systems and their operational consequences. It also analyses diagnostic capabilities of the system in order to support preventive and corrective maintenance actions.

5.5 PRELIMINARY RELIABILITY ANALYSIS

Upon completion of the RAM Plan, it is recommended to create a preliminary Reliability Analysis document. This document must provide a rough estimation of the availability and reliability of the sub-systems described in chapter 3.

All relevant documentation shall be reviewed in order to define, at a preliminary level, the overall system structure and to recognise all the functional requirements, which may affect the RAM performance of the system.

5.6 RELIABILITY ANALYSIS

The Reliability Analysis is a key document, with focus on HW relevant aspects, describing reliability, availability and maintainability features of Integrated Platform Services (IPS) solution element – IPS Solution OBB 1.0.0 (IPSS OBB 1.0.0).

Reliability aspects of SW have to be assured by following the rules of [EN 50126-1]. An analysis of SW reliability is not part of the Reliability Analysis where only permanent random HW faults and system failures will be analysed.

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Failure rates for indoor components will be calculated

To be able to make reliable statements and to demonstrate that the required RAM goals will be obtained, the following information about the hardware and the purpose of the system has to be collected:

- what kind of sub-systems / components will be used,
- quantity of each sub-system / component,
- Failure rates of all PBAs and devices used,
- MTBF and MTTR for each sub-system / component.

The Reliability Analysis shall point out measures, how availability can be reached according to the mission goals.

5.7 MAINTAINABILITY

To ensure a proper installation, operation and maintenance of the system, corresponding installation and maintenance documents have to be provided. MTTR and related terms have to be checked with respect to fulfilment of information stated in these documents.

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N/A	562	16282~9950	01 RL
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6. RAM DELIVERABLES

6.1 LIST OF RAM DELIVERABLE DOCUMENTS

This section summarizes the documents which are foreseen during RAM activities:

- RAM Plan (this document),
- preliminary Reliability Analysis (recommended),
- RAM HW-FMEA (recommended),
- Reliability Analysis (required).

Note: RAM documentation must be available according to the Document List and Review Matrix for engineering projects [DLRM].

6.2 RAM REQUIREMENTS

The RAM related customer requirements need to be checked, clarified if necessary and appropriately refined during the different development phases. If additional RAM related aspects are to be considered in order to fulfill RAM related customer requirements, they should comply with corresponding standards [EN 50126-1] and the definitions defined in [RMP_GA] and [RMP_IPS].

The Solution Element – IPS Solution OBB 1.0.0 (IPSS OBB 1.0.0) is subject to the **DPR** requirements stated in [SSS_IPS] (see Table 7 below) and furthermore in [PH_RAMs].

ID	Synopsis
EL4-IPS-SSS-REQ-00010-DPR	[IPS] shall provide local redundant centralized alerting capabilities for infrastructure, networks and applications
EL4-IPS-SSS-REQ-00013-DPR	[IPS] shall provide highly available geo-redundant setup for all infrastructure and services including compute, storage and network, across up to 3 geographically separated DC locations
EL4-IPS-SSS-REQ-00014-DPR	[IPS] shall provide multiple, dedicated virtual networks with configurable Quality of Service (QoS) parameters. These networks shall be managed as Software-defined-Network (SDN) enabling centralized control, dynamic configuration and seamless scalability
EL4-IPS-SSS-REQ-00021-DPR	[IPS] shall provide automatic switch-over to geo-redundant connections in the event of primary connection failure
EL4-IPS-SSS-REQ-00022-DPR	[IPS] shall provide manual switch-over capabilities to geo-redundant location, allowing to re-route network traffic in the event of primary connection failure
EL4-IPS-SSS-REQ-00024-DPR	[IPS] shall support geo-redundant, consistent configuration and application binary storage. Configurations, VMs and Images shall be synchronized between Datacenters
EL4-IPS-SSS-REQ-00025-DPR	[IPS] shall support local, consistent configuration and application binary storage. Configurations, VMs and Images shall be locally available
EL4-IPS-SSS-REQ-04009-DPR	The availability of the database systems within the IPS INFRA and SERVICE block shall be 99.9999%.
EL4-IPS-SSS-REQ-01102-DPR	The Core Network shall be able to provide two-fold redundant connections to the Power Supply Monitor device from Gustav Klein

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ID	Synopsis
EL4-IPS-SSS-REQ-01103-DPR	The Core Network shall be able to provide two-fold redundant connections to the OEGB network Datennetz 2020
EL4-IPS-SSS-REQ-01301-DPR	The Core Network shall provide 99.9999% availability
EL4-IPS-SSS-REQ-01304-DPR	The network connection to each server cabinet shall be two-fold redundant
EL4-IPS-SSS-REQ-01306-DPR	The network connection to each RaSTA network shall be two-fold redundant
EL4-IPS-SSS-REQ-02001-DPR	The volumes provided by the storage system shall provide 99.9999% availability.
EL4-IPS-SSS-REQ-02002-DPR	The operating system on all storage servers shall use at least 2 storage devices in a RAID1 array.
EL4-IPS-SSS-REQ-02003-DPR	The storage server shall provide additional storage devices for the provision of the common storage space for the applications.
EL4-IPS-SSS-REQ-03101-DPR	The operating system on all compute servers shall use at least 2 storage devices in a RAID1 array.
EL4-IPS-SSS-REQ-04015-DPR	[IPS] shall provide high available storage and computing services spanning 3 availability zones
EL4-IPS-SSS-REQ-04025-DPR	[IPS] shall support local Availability Zone concept for VMs
EL4-IPS-SSS-REQ-04026-DPR	[IPS] shall support Geo-Redundancy (multiple regions) concept for VMs
EL4-IPS-SSS-REQ-04030-DPR	[IPS] shall support automatic restart of the VMs on the server after this server restart
EL4-IPS-SSS-REQ-04031-DPR	[IPS] shall support scale-in / out (increase or decrease) additional computing resources per Availability zone
EL4-IPS-SSS-REQ-04032-DPR	[IPS] shall support scale-in / out (increase or decrease) additional network resources per Availability zone
EL4-IPS-SSS-REQ-04033-DPR	[IPS] shall support scale-in / out (increase or decrease) additional storage resources per Availability zone
EL4-IPS-SSS-REQ-05104-DPR	The INFRA cluster shall provide a storage for application files with an availability of 99.9999%.
EL4-IPS-SSS-REQ-05105-DPR	The INFRA cluster shall provide a storage for data preparation files with an availability of 99.9999%
EL4-IPS-SSS-REQ-05106-DPR	The INFRA cluster shall provide a storage for configuration files with an availability of 99.9999%
EL4-IPS-SSS-REQ-05306-DPR	The availability of the PKI service for management of certificates shall be 99.9999%.
EL4-IPS-SSS-REQ-06306-DPR	The availability of the message broker shall be 99.9999%.
EL4-IPS-SSS-REQ-08306-DPR	IPS shall provide redundant communication path between NMR application instances.

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ID	Synopsis
EL4-IPS-SSS-REQ-09902-DPR	IPS shall redistribute remaining services following the balanced increment principle in case of a server failure
EL4-IPS-SSS-REQ-09903-DPR	IPS shall not redistribute services in case of server reintegration to prevent any additional disruption of operating services.
EL4-IPS-SSS-REQ-09906-DPR	IPS shall execute a group of services which belongs to a specific sub-station on the same physical server (colocation/grouping).
EL4-IPS-SSS-REQ-09911-DPR	IPS shall place IPS service instances belonging to the IPS INFRA Cluster on different servers and different availability zones following the balanced distribution principle.
EL4-IPS-SSS-REQ-09912-DPR	IPS shall execute IPS service instances belonging to the IPS INFRA Cluster 2oo3 hot-redundant or 1oo2 hot-redundant or 1oo3 hot-redundant (high availability).
EL4-IPS-SSS-REQ-09913-DPR	IPS shall place IPS Service instances belonging to an IPS SERVICE Cluster on different servers and different availability zones following the balanced distribution principle.
EL4-IPS-SSS-REQ-09914-DPR	IPS shall execute IPS Service instances belonging to an IPS SERVICE Cluster 2oo3 hot-redundant or 1oo2 hot-redundant or 1oo3 hot-redundant (high availability).

Table 7: IPS DPR requirements

6.3 PERIODICAL RAM ACTIVITIES REPORT

The RAM Plan can be updated if necessary, in each phase of the product life cycle, described in [EN 50126-1].

Possible reasons are:

- Review comments of this document,
- Hardware changes,
- Change of standards and procedures.

Each change of the RAM Plan results in a new edition. At each new edition, the reasons for change will be documented in section 1.5

END OF DOCUMENT

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